In each case, the first half of the state-label describes the state of Alice's coin, and the second half describes the state of Bob's die.

Sometimes, we'll need to refer to an arbitrary basis vector in S_{AB}. To do that, we'll use ket-vectors that look like this,

$$|ab\rangle,$$

or like this,

$$|a'b'\rangle.$$

In this notation, the *a* or *a′* (or whatever the left-hand character of the label happens to be) represents one of Alice's states, and the *b* or *b′* represents one of Bob's states.

There is one aspect of this notation that is tricky. Even though our S_{AB} state-labels are doubly indexed, ket-vectors like |*ab*> or |*H*3> represent a *single state* of the combined system. In other words, we're using a double index to label a single state. This will take some getting used to. Alice's part of the state-label is always on the left and Bob's part is always on the right—keeping Alice and Bob in alphabetical order makes this convention easy to remember.

The rules are the same for more general systems. The only difference is that the two A-states and the six B-states would be replaced by N_A and N_B states respectively, and the tensor product would have dimension

$$N_{AB} = N_A N_B.$$

Systems with three or more components can be represented by tensor products of three or more state spaces, but we won't do that here.

Now that we've described Alice's and Bob's separate spaces S_A and S_B, as well as the combined space S_{AB}, there's still one more bit of notation to set up. Alice has a set of operators, labeled σ, that act on her system. Bob has a